University of BRISTOL

COMS20018, Introduction to AI:
AI Ethics II:
Measuring and Achieving Fairness

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3rd November 2026

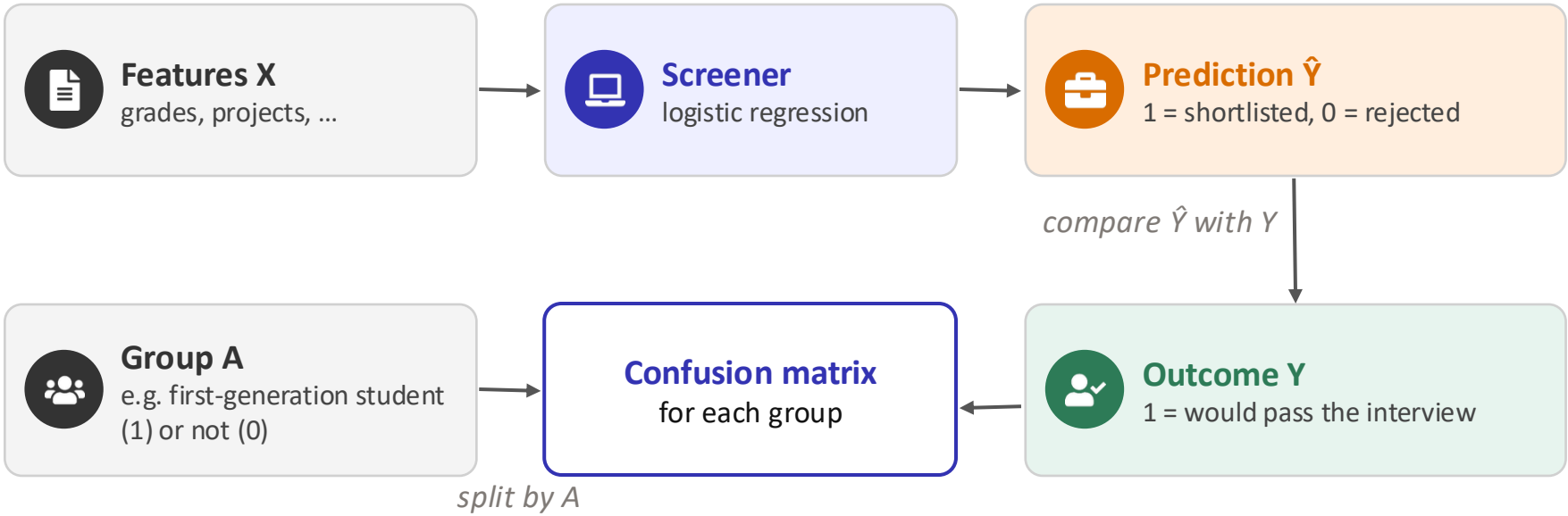
Agenda

1. **Recap & groups**
   X, A, Y, Ŷ; what is a group?
2. **Metrics by group**
   the same model, split
3. **Fairness criteria**
   parity, opportunity, odds…
4. **You can't have it all**
   why criteria conflict
5. **The cost of fairness**
   accuracy vs fairness
6. **Achieving fairness**
   pre-, in-, post-processing

Today's question

What does it mean for a classifier to be fair, and how can we measure it and make it fairer?

Recap from L11: the screener and its notation



Three ideas from L11 we build on today

Metrics

TPR, FPR, precision, selection rate

Bias

data, labels and proxies

Explanations

LIME / SHAP found a proxy for gender

Running example: the same screener and the same 200 test applicants as L11 (accuracy 0.810).

Before fairness: what is a group?

Fairness compares **groups** defined by a sensitive attribute A. UK law lists nine protected characteristics:

age · disability · gender reassignment · marriage & civil partnership · pregnancy & maternity · race · religion or belief · sex · sexual orientation

1 Groups are constructed

Categories can be ambiguous and contested.

Example

Who is "first-generation"? Neither parent has a degree, or just one? Each definition gives a different group.

2 People belong to many groups

Intersections matter.

Example

Face-analysis systems: error up to 34.7% for darker-skinned women, at most 0.8% for lighter-skinned men.

3 Group fairness ≠ individual fairness

Fair on average can still be unfair to a person.

Example

Both groups shortlisted at 50%, yet a strong applicant is rejected while a weaker one in the same group is shortlisted.

Today: $A \in \{0, 1\}$ (binary for simplicity), e.g. first-generation student or not.

Sources: Menon & Williamson (2018), Appendix H · UK Equality Act 2010, s.4 · Buolamwini & Gebru (2018), Gender Shades

A question to begin with: is the screener fair?

Overall it looks fine. Now split the same 200 applicants by group:

Group A = 0 (120 applicants)		
	$\hat{Y} = 1$	$\hat{Y} = 0$
Y = 1	48	12
Y = 0	12	48

Group A = 1 (80 applicants)		
	$\hat{Y} = 1$	$\hat{Y} = 0$
Y = 1	12	8
Y = 0	6	54

Check: the two tables add up to L11's matrix (TP 60, FN 20, FP 18, TN 102).

Questions to keep in mind

1. Is the screener equally accurate for both groups?
2. Are both groups shortlisted at the same rate? Should they be?
3. Is a capable student equally likely to be shortlisted in both groups?
4. Does a shortlist decision mean the same thing for both groups?

Measuring performance by group

Same formulas as L11, computed only on the applicants with A = a

Quantity	Probability	From group a's matrix	Meaning for group a
Base rate	$P(Y = 1 \mid A = a)$	$\frac{\mathrm{TP} + \mathrm{FN}}{N_a}$	fraction who would pass
Selection rate	$P(\hat{Y} = 1 \mid A = a)$	$\frac{\mathrm{TP} + \mathrm{FP}}{N_a}$	fraction shortlisted
TPR	$P(\hat{Y} = 1 \mid Y = 1, A = a)$	$\frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$	a capable student is shortlisted
FPR	$P(\hat{Y} = 1 \mid Y = 0, A = a)$	$\frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}$	a not-yet-ready student is shortlisted
Precision (PPV)	$P(Y = 1 \mid \hat{Y} = 1, A = a)$	$\frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$	a shortlisted student would pass

Notations

- N_a: number of applicants in group a
- PPV: positive predictive value: how much a positive prediction $\hat{Y} = 1$ can be trusted
- $P(\text{event} \mid \text{condition})$: probability of the event among applicants who satisfy the condition

Try it: the screener, group by group

Group A = 0		
	$\hat{Y} = 1$	$\hat{Y} = 0$
Y = 1	48	12
Y = 0	12	48

Group A = 1		
	$\hat{Y} = 1$	$\hat{Y} = 0$
Y = 1	12	8
Y = 0	6	54

Try it (5 min)

1. For each group, calculate $\text{TPR} = \frac{\text{TP}}{\text{TP+FN}}$ and $\text{FPR} = \frac{\text{FP}}{\text{FP+TN}}$.

2. For each group, calculate the selection rate and the precision.

3. For each group, calculate the accuracy. Is the screener equally accurate for both groups?

4. Which group does the screener treat worse, and in what sense?

Try it: the screener, group by group (answer)

	A = 0	A = 1	Gap
TPR	$48/60 = 0.80$	$12/20 = 0.60$	**0.20**
FPR	$12/60 = 0.20$	$6/60 = 0.10$	**0.10**
Selection rate	$60/120 = 0.50$	$18/80 = 0.225$	**0.275**
Precision	$48/60 = 0.80$	$12/18 \approx 0.667$	**0.133**
Accuracy	$96/120 = 0.800$	$66/80 = 0.825$	−0.025
Base rate	$60/120 = 0.50$	$20/80 = 0.25$	0.25

3–4. Accuracy is slightly higher for A = 1, but capable A = 1 students are shortlisted less often (TPR 0.60 vs 0.80), and A = 1 is shortlisted at less than half the rate.

What the average hid

Equal accuracy, yet a capable A = 1 student is rejected twice as often (FNR 0.40 vs 0.20). Which gap matters is a choice: that is what fairness criteria formalise.

Demographic parity (and disparate impact)

$$P(\hat{Y} = 1 \mid A = 0) = P(\hat{Y} = 1 \mid A = 1)$$

Both groups are shortlisted at the same rate.

Our screener

Selection rates 0.50 vs 0.225.

$$\text{DI} = \frac{P(\hat{Y} = 1 \mid A = 1)}{P(\hat{Y} = 1 \mid A = 0)} = \frac{0.225}{0.50} = 0.45$$

✗ **Not satisfied (and DI < 0.8)**

Makes sense when...

- Historical labels are untrustworthy
- Representation is itself the goal

Limitations

- Ignores Y: a perfect screener fails it when base rates differ
- Can be met by shortlisting at random

Notation

- A: the group; $\hat{Y}$: prediction; $P(\hat{Y} = 1 \mid A = a)$: selection rate of group a
- DI: disparate-impact ratio (the group with the lower rate on top)

Sources: Barocas, Hardt & Narayanan, fairmlbook.org · US EEOC four-fifths rule (29 CFR 1607.4)

Equal opportunity

$$P(\hat{Y} = 1 \mid Y = 1, A = 0) = P(\hat{Y} = 1 \mid Y = 1, A = 1)$$

Among students who would pass, both groups have the same chance of being shortlisted.

Our screener

TPR 0.80 vs 0.60.

Capable A = 1 students lose out.

✗ **Not satisfied: TPR gap 0.20**

Makes sense when…

- The positive decision is a benefit
- Wrongly denying it is the main harm

Limitations

- Says nothing about false positives
- Trusts the labels Y

Notation

- A: the group (0 or 1); Y: observed outcome; $\hat{Y}$: prediction
- $P(\text{event} \mid \text{condition})$: probability of the event among applicants who satisfy the condition

Source: Hardt, Price & Srebro (2016)

Equalised odds

$$P(\hat{Y} = 1 \mid Y = y, A = 0) = P(\hat{Y} = 1 \mid Y = y, A = 1) \quad \text{for } y \in \{0,1\}$$

Equal TPR (y = 1) and equal FPR (y = 0) across groups: both kinds of mistake are shared equally.

Our screener

TPR 0.80 vs 0.60 ✗

FPR 0.20 vs 0.10 ✗

✗ **Not satisfied: both rates differ**

Makes sense when…

- Both errors cause real harm
- Labels are trustworthy

Limitations

- Strong: may need randomised decisions
- Also trusts the labels Y

Notation

- A: the group; Y: observed outcome; $\hat{Y}$: prediction
- y: either outcome value, so this is two conditions: $y = 1$ gives TPR, $y = 0$ gives FPR

Source: Hardt, Price & Srebro (2016)

Predictive parity

$$P(Y = 1 \mid \hat{Y} = 1, A = 0) = P(Y = 1 \mid \hat{Y} = 1, A = 1)$$

A shortlist decision means the same for every group: equal precision (PPV).

Our screener

Precision 0.80 vs 0.667.

A shortlist is a weaker signal for A = 1.

✗ **Not satisfied: gap 0.133**

Makes sense when…

- People act on the prediction itself
- e.g. interviewers reading a shortlist

Limitations

- Conditions on the prediction, not the truth
- Can hold while error rates differ widely

Notation

- A: the group (0 or 1); Y: observed outcome; $\hat{Y}$: prediction
- $P(\text{event} \mid \text{condition})$: probability of the event among applicants who satisfy the condition

Source: Chouldechova (2017)

Fairness criteria at a glance

Criterion	Requires equal…	Part of the matrix	Uses Y?	Screener
Demographic parity	selection rate	whole group	No	0.50 vs 0.225 ✗
Disparate impact	ratio of selection rates ≥ 0.8	whole group	No	DI = 0.45 ✗
Equal opportunity	TPR	Y = 1 row	Yes	0.80 vs 0.60 ✗
Equalised odds	TPR and FPR	both rows	Yes	both differ ✗
Predictive parity	precision	$\hat{Y}$ = 1 column	Yes	0.80 vs 0.667 ✗

Three families

Independence

$\hat{Y} \perp A$

decisions do not depend on the group

Separation

$\hat{Y} \perp A \mid Y$

errors do not depend on the group

Sufficiency

$Y \perp A \mid \hat{Y}$

a prediction means the same for all groups

Notation

$\perp$: “is independent of”; $\mid Y$: “within each value of Y”

Source: Barocas, Hardt & Narayanan, fairmlbook.org

Try it: which fairness criteria hold?

A company shortlists students for a coding-bootcamp scholarship (Y = 1: would complete the bootcamp; $\hat{Y} = 1$: shortlisted). It is evaluated on 100 students from each group.

Group A = 0		
	$\hat{Y} = 1$	**$\hat{Y} = 0$**
Y = 1	40	10
Y = 0	10	40

Group A = 1		
	$\hat{Y} = 1$	**$\hat{Y} = 0$**
Y = 1	16	4
Y = 0	16	64

Try it (5 min)

1. For each group, calculate the selection rate, TPR, FPR and precision.

2. Which of these does the classifier satisfy: demographic parity, equal opportunity, equalised odds, predictive parity?

3. Is anything surprising about your answer to 2?

Try it: which fairness criteria hold? (answer)

	A = 0	A = 1
Selection rate	$\frac{40+10}{100} = 0.50$	$\frac{16+16}{100} = 0.32$
TPR	$\frac{40}{40+10} = 0.80$	$\frac{16}{16+4} = 0.80$
FPR	$\frac{10}{10+40} = 0.20$	$\frac{16}{16+64} = 0.20$
Precision	$\frac{40}{40+10} = 0.80$	$\frac{16}{16+16} = 0.50$

✗ Demographic parity

✓ Equal opportunity

✓ Equalised odds

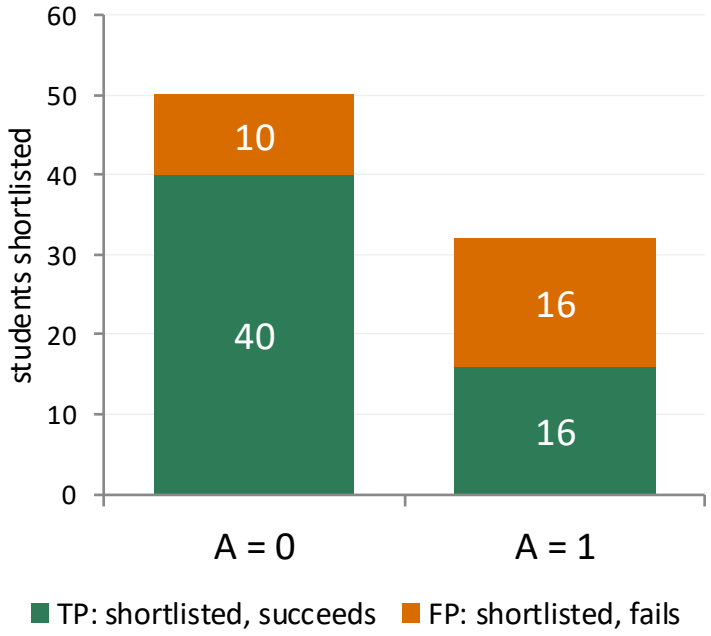
✗ Predictive parity

3. The surprise

Both groups get exactly the same error rates (TPR 0.80, FPR 0.20), yet a shortlist means very different things: 80% of shortlisted A = 0 students would complete, but only 50% of A = 1 students. The base rates differ (50 vs 20 of 100). Next: why this must happen.

Why we cannot have everything

In the Try it, both groups had the same TPR (0.80) and FPR (0.20). So why is precision so different?



$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

A = 0: 50 succeed, 50 fail

$$\text{TP} = 40,\ \text{FP} = 10 \ \Rightarrow\ \frac{40}{50} = 0.80$$

A = 1: 20 succeed, 80 fail

$$\text{TP} = 16,\ \text{FP} = 16 \ \Rightarrow\ \frac{16}{32} = 0.50$$

Different base rates ⇒ equal error rates and equal precision cannot both hold.

Sources: Chouldechova (2017) · Kleinberg et al. (2017)

Comparing criteria: outcomes or errors?

Outcome-based

Who receives the benefit?

	$\hat{Y} = 1$	$\hat{Y} = 0$
$Y = 1$	**TP**	FN
$Y = 0$	**FP**	TN

compares who is shortlisted ($\hat{Y} = 1$)

Demographic parity, disparate impact

✓ No labels Y needed

✗ Ignores who would actually succeed

Error-based

Who bears the mistakes?

	$\hat{Y} = 1$	$\hat{Y} = 0$
$Y = 1$	TP	**FN**
$Y = 0$	**FP**	TN

compares mistakes, using the true Y

Equal opportunity, equalised odds, predictive parity

✓ Fits when mistakes cause the harm

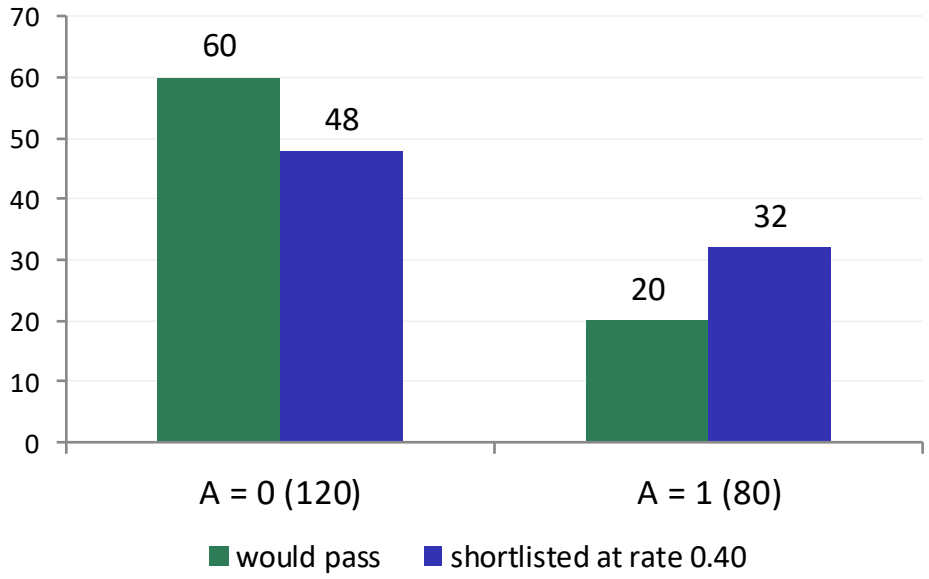
✗ Trusts Y, so inherits label bias

Choosing a criterion is a value judgement, not a technical default.

Source: Barocas, Hardt & Narayanan, fairmlbook.org

Worked example: the cost of fairness

Take a perfect screener: it shortlists exactly those who would pass (60 of 120 in A = 0, 20 of 80 in A = 1). Its rates are 0.50 vs 0.25, so parity fails. Now force both groups to shortlist at the same rate 0.40:



A = 0: 48 places for 60 who would pass → **12 capable students rejected**

A = 1: 32 places for 20 who would pass → **12 not-ready students shortlisted**

24 mistakes

accuracy 100% → 88%

Forcing fairness can cost accuracy, even for a perfect model.

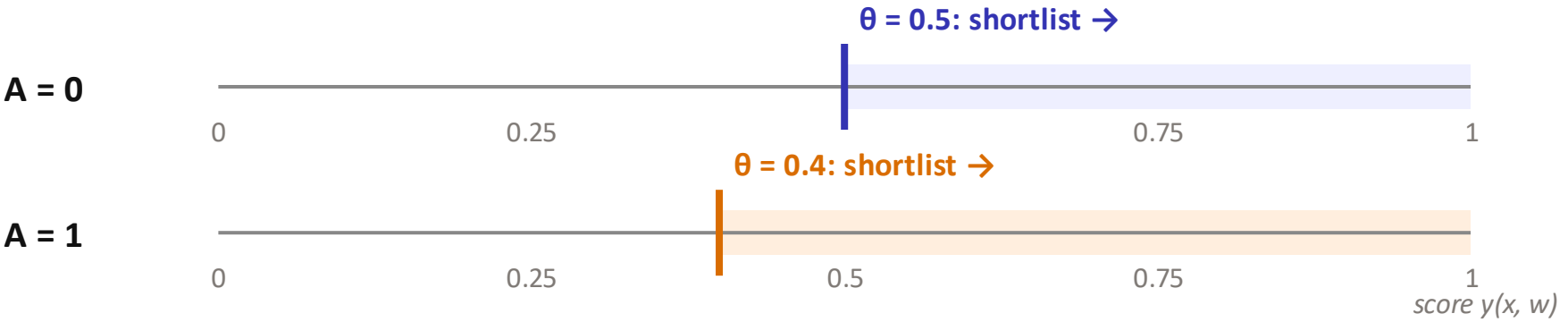
Source: Menon & Williamson (2018)

The most accurate fair classifier: move the threshold

Recall (L03): the screener computes a score $y(\mathbf{x}, \mathbf{w})$ (a probability) and shortlists if $y(\mathbf{x}, \mathbf{w}) > \theta$. Without fairness constraints, θ = 0.5 gives the highest accuracy.

Question: among all classifiers that satisfy a fairness criterion, which one makes the fewest mistakes?

Answer (Menon & Williamson, 2018): keep the same score, just move the threshold. If the group A is known, use one threshold per group: shortlist if $y(\mathbf{x}, \mathbf{w}) > \theta_A$.

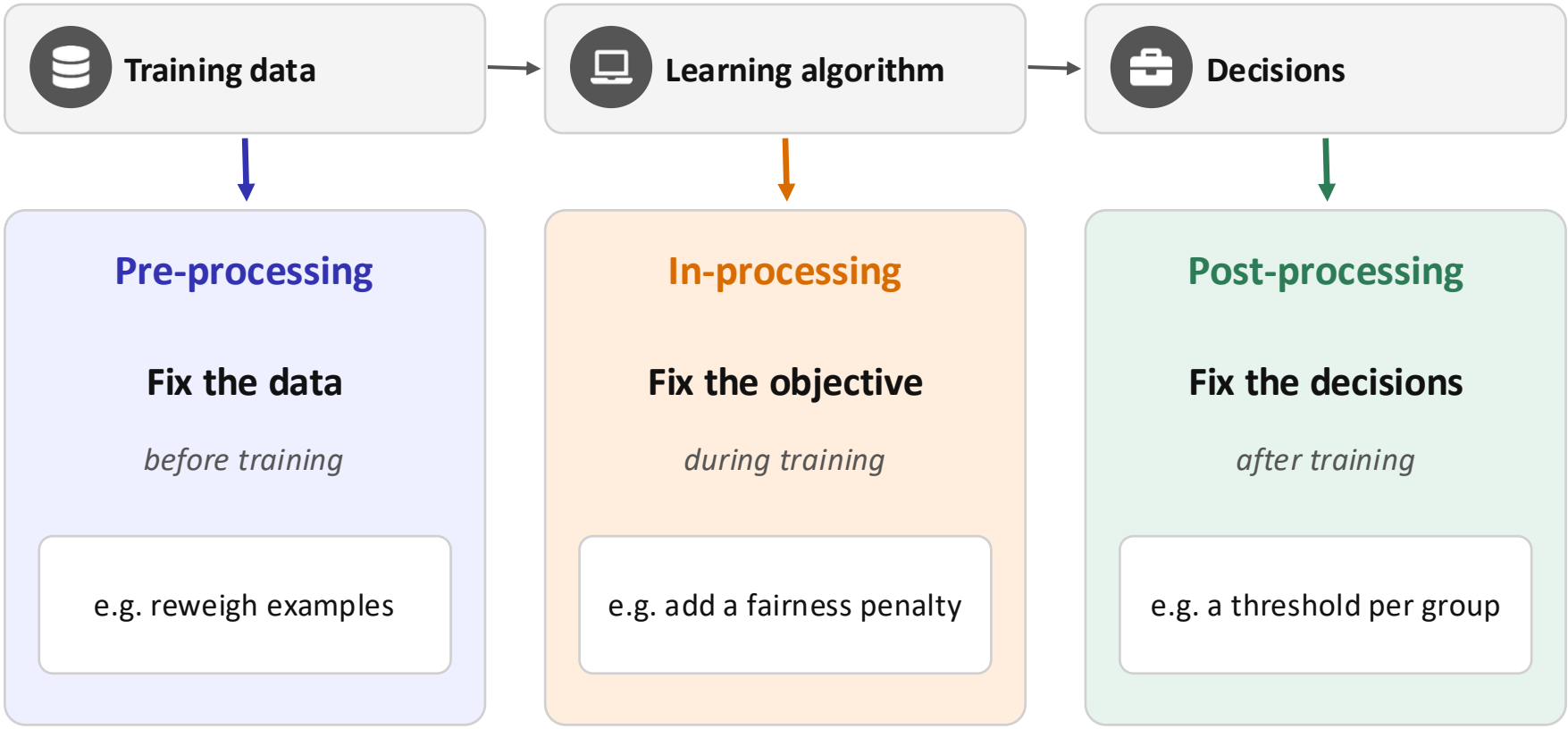


Notation

θ_A: threshold for group A, i.e. θ_0 for A = 0 and θ_1 for A = 1

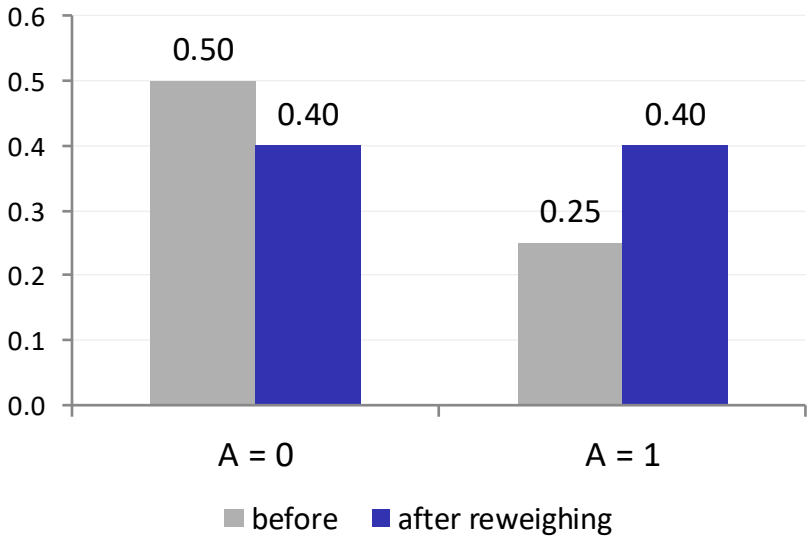
Source: Menon & Williamson (2018)

Achieving fairness: three places to intervene



Pre-processing: reweighing the training data

Idea: give rare combinations (e.g. capable A = 1 students) more weight, so the training data no longer link group and outcome. Then train as usual.



Weight for each kind of student

0.8 A = 0, would pass	**1.2** A = 0, would not
1.6 A = 1, would pass	**0.8** A = 1, would not

$$w(a,y) = \frac{P(A=a)\,P(Y=y)}{P(A=a,\,Y=y)}$$

Notation

- $w(a,y)$: weight of each example in group a with outcome y
- $P(A=a,\,Y=y)$: fraction of data with both
- Base rate of group a: $P(Y=1 \mid A=a)$, the fraction who would pass

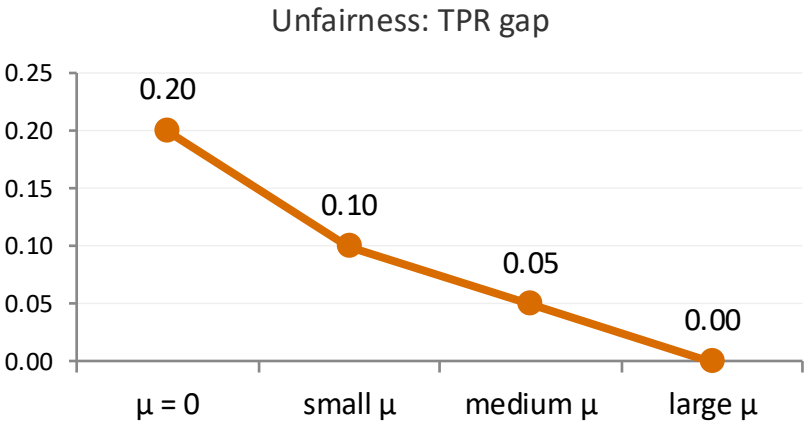
Source: Kamiran & Calders (2012)

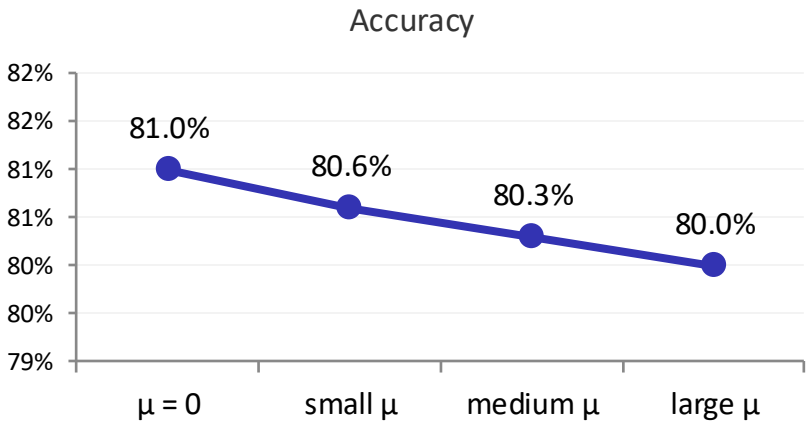
In-processing: build fairness into training

Idea: train to make few mistakes and small group gaps at the same time.

$$\min_{\mathbf{w}} \; L(\mathbf{w}) \; + \; \mu \cdot \text{Unfairness}(\mathbf{w})$$

usual loss (L03) *group gap, weighted by μ*





Larger μ: smaller gap, slightly lower accuracy (illustrative). Tools: Fairlearn reductions, adversarial debiasing.

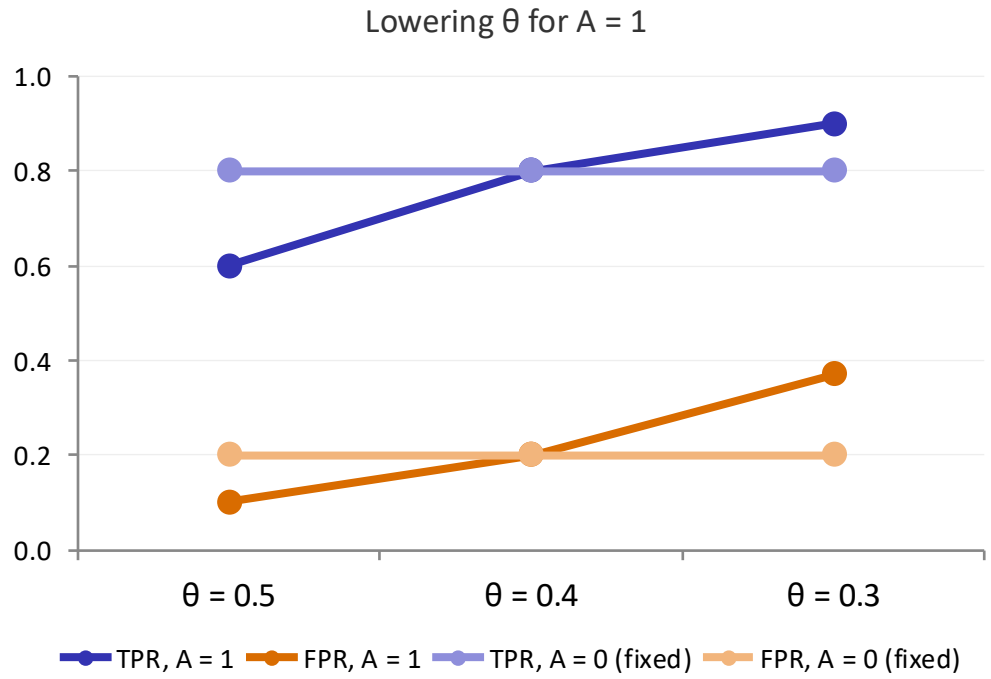
Notation

- $\mathbf{w}$: model weights; $L(\mathbf{w})$: training loss (cross-entropy); $\mu \geq 0$: how much fairness matters
- $\text{Unfairness}(\mathbf{w})$, e.g. the TPR gap $|\text{TPR}_0 - \text{TPR}_1|$; TPR = fraction of capable students shortlisted

Sources: Zafar et al. (2017) · Agarwal et al. (2018) · Zhang et al. (2018)

Post-processing: group-specific thresholds

Idea: keep the trained model. Keep θ = 0.5 for A = 0 and lower θ for A = 1 until the rates match.



θ = 0.5 (same for both)

TPR 0.60 vs 0.80: unfair to capable A = 1

accuracy 81%

θ = 0.4 ✓ equalised odds

TPR and FPR now equal across groups

accuracy 80%

θ = 0.3 ✓ demographic parity

both groups shortlisted at rate 0.50

accuracy 76%

Notation

- θ: shortlist if the score $y(\mathbf{x}, \mathbf{w}) > \theta$; TPR: fraction of capable students shortlisted
- FPR: fraction of not-ready students shortlisted. Needs A at decision time, but no retraining

Source: Hardt, Price & Srebro (2016)

Comparing the three approaches

	Pre-processing	In-processing	Post-processing
What changes?	the data	the training objective	the threshold(s)
Retrain the model?	yes	yes	no
Works with any model?	yes	no: must change training	yes, even a black box
Needs A when deciding?	no	no	yes

green = advantage, red = drawback

In practice: open-source toolkits such as Fairlearn and AI Fairness 360 implement all three.

Sources: Fairlearn · AI Fairness 360

Fairness is socio-technical, not just a metric

1 Start with the problem

Should this decision be automated at all? Who could be harmed?

2 Question the labels

If Y encodes past discrimination, equalising errors on Y is not enough.

3 Choose criteria openly

With the people affected; document why others were rejected.

4 Report per group

Including intersections, e.g. in model cards (Mitchell et al., 2019).

5 Explain decisions

LIME / SHAP to find proxies, and to let people contest decisions (L11).

6 Monitor and stay accountable

Shift and feedback loops can undo fairness; keep redress open.

Source: Mitchell et al. (2019), Model Cards

Summary

Five things to remember

1. Fairness compares groups, and choosing the groups is itself a decision.

2. Demographic parity: selection rate. Equal opportunity: TPR. Equalised odds: TPR and FPR. Predictive parity: precision. Read the condition to find the row or column.

3. With different base rates, equal error rates and equal precision cannot both hold.

4. Fairness constraints can cost accuracy; the most accurate fair classifier just moves the threshold, one per group.

5. Interventions: pre-, in- and post-processing, each with trade-offs. Choosing the criterion is a value judgement.

Reading

Core: Fairness and Machine Learning, Ch. 3 (Classification)
Barocas, Hardt & Narayanan (2023). Independence, separation, sufficiency.

The cost of fairness in binary classification
Menon & Williamson (2018), FAT*, PMLR 81. Appendix H discusses groups.

Equality of opportunity in supervised learning
Hardt, Price & Srebro (2016), NeurIPS. Equal opportunity, equalised odds, post-processing.

Fair prediction with disparate impact
Chouldechova (2017), Big Data. The impossibility result and COMPAS.

Fairlearn user guide
Metrics and mitigation algorithms in Python.